

NIEB HASAN NEOM

Rolla, MO

✉ niebhasanneom0@gmail.com

🌐 linkedin.com/in/nieb-hasan-neom

🐙 github.com/NiebHasan077

🎓 [Google Scholar](#)

SUMMARY

Computer Science Ph.D. student and **Kummer Doctoral Fellow** at Missouri S&T researching **agentic optimization of large-scale scientific data movement**. **Four peer-reviewed publications**, **266 citations**, including first-author work at **WORKS26 @ SC'26** and a co-authored paper at **IEEE eScience 2026**, alongside three years of backend engineering on a 30-microservice platform. Interests: HPC data systems, distributed infrastructure, LLM workflow orchestration.

EDUCATION

Missouri University of Science & Technology

Rolla, MO, USA

Doctor of Philosophy in Computer Science

Aug. 2025 – Present

- **Kummer Innovation and Entrepreneurship Doctoral Fellow** | **GPA: 4.00/4.00** | Anticipated graduation: **05/2030**

Khulna University of Engineering & Technology

Khulna, Bangladesh

Bachelor of Science in Computer Science and Engineering

Feb. 2017 – April 2022

- **CGPA: 3.51/4.00** | KUET Technical Scholarship (2017–2022)

RESEARCH EXPERIENCE

Computer Systems & Networking Lab, Missouri University of Science & Technology

Rolla, MO

Graduate Research Assistant | Advisor: Dr. Md Arifuzzaman

Aug. 2025 – Present

- Designed a **guarded, state-aware orchestration** layer extending LLM function calling with workflow semantics—multi-turn input accumulation, typed-schema argument constraints, blocked execution on incomplete state, and correction/what-if/reset steering—to give domain scientists natural-language access to trained prediction models (**accepted, WORKS26 @ SC'26**)
- Instantiated the workflow for protein-corona prediction over a backend of **178 per-protein classifiers (87.13% AUC-ROC)**; measured **98.5% feature-extraction accuracy** across 300 cases and **300/300** passing reliability scenarios spanning 50 nanoparticle configurations, in collaboration with Louisiana State University
- Building an **LLM-driven data-movement optimizer** that tunes network transfer concurrency from live TCP telemetry (congestion window, retransmissions, RTT variance) via a two-agent Diagnostic/Decision pipeline on an OBSERVE–REASON–ACT–EVALUATE loop, with zero-copy `sendfile()` workers, bounded-delta caps and oscillation damping
- Co-authored self-tuning segmented retrieval for large-scale genomic data (**accepted, IEEE eScience 2026**); evaluating transfer systems on the **CloudLab**, **FABRIC** and **TACC/ACCESS** testbeds

INDUSTRY EXPERIENCE

Kona Software Lab Limited

Gulshan, Dhaka, Bangladesh

Software Engineer II (promoted from Graduate Engineer, 2024)

Apr. 2022 – Jun. 2025

- Backend engineer on **Kona Token Trade**, a cryptocurrency and blockchain-based token trading platform; owned **6** of the platform's **30 Spring Boot microservices** and contributed to **15** in total, covering trade, wallet, user, payment and fee management
- Replaced synchronous inter-service REST calls with **Kafka** messaging, closing a fault-tolerance gap in which a single unavailable downstream service could fail an in-flight transaction; introduced caching that cut system response time by **30%**
- Implemented **two-factor authentication** on critical financial operations such as asset buy and sell, and integrated bank-account and KONA card payment gateways
- Built the **Card Personalization System** (Apache Struts, Java), completing the smart-card personalization pipeline against **GlobalPlatform**, **Visa**, **MasterCard** and **EMV** specifications, now in commercial use; selected from the engineering team for a **14-day training program** on the project at parent company Kona I in **Seoul, South Korea**
- Built against **AWS** (S3, SNS, EC2) from the backend services; mentored and onboarded **2 junior engineers** into backend engineering fundamentals

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, Bash

AI & Agentic Systems: LLM function calling and tool use, schema-constrained generation, agentic workflow orchestration, Ollama, fine-tuned HuggingFace models, OpenAI API, Google Gemini API, PyTorch, Pydantic, NumPy

HPC & Systems: CloudLab, FABRIC, TACC/ACCESS, Falcon data mover, TCP performance analysis, multiprocessing, zero-copy `sendfile()`, Linux systems programming

Distributed Systems & Backend: Spring Boot, Apache Kafka, microservice architecture, RESTful web services, AWS (S3, SNS, EC2)

Databases: PostgreSQL, MongoDB, Redis, MySQL

Developer Tools: Git, Gradle, Jenkins, Jira, IntelliJ IDEA, DataGrip, VS Code, LaTeX

PUBLICATIONS

- **Nieb Hasan Neom**, Manas Ranjan Gartia, and Md Arifuzzaman. “Guarded, State-Aware Orchestration for Schema-Constrained Scientific Model Execution.” *21st Workshop on Workflows in Support of Large-Scale Science (WORKS26)*, held in conjunction with *SC26*, Chicago, IL, Nov. 2026. (**Accepted — to appear**)
- Rasman Muhtasim Swargo, **Nieb Hasan Neom**, and Md Arifuzzaman. “Self-Tuning Segmented Retrieval of Large-Scale Genomic Data.” *22nd IEEE International Conference on e-Science (eScience 2026)*, Naples, Italy, Sept. 2026. (**Accepted — to appear**)
- Sunanda Das, Md Samir Imtiaz, **Nieb Hasan Neom**, Nazmul Siddique, and Hui Wang. “A hybrid approach for Bangla sign language recognition using deep transfer learning model with random forest classifier.” *Expert Systems with Applications* 213 (2023): 118914. **Impact Factor: 8.665 | 218 citations.** [Link](#)
- Md Milon Islam, **Nieb Neom**, Md Samir Imtiaz, Sheikh Nooruddin, Md Repon Islam, and Md Rabiul Islam. “A Review on Fall Detection Systems Using Data from Smartphone Sensors.” *Ingénierie des Systèmes d’Information* 24, no. 6 (2019): 569–576. **48 citations.** [Link](#)

PROJECTS

LLM-based Data Movement Optimizer | *Python, Ollama, HuggingFace, Pydantic, NumPy, Linux* **2026**

- Built an LLM-driven file transfer system that adapts network concurrency in real time, replacing the hand-tuned static configuration conventional movers require; sender/receiver architecture over TCP with a multiprocessing worker pool and zero-copy `sendfile()` transfers
- Implemented a two-agent Diagnostic/Decision pipeline with structured memory and journal logging, plus execution safety: bounded per-decision deltas, oscillation detection and damping, graduated ramping of large changes, and hard concurrency limits
- Collects live TCP socket statistics, RTT variance, retransmissions and system load each probing interval to drive an OBSERVE–REASON–ACT–EVALUATE feedback loop; pluggable across local Ollama models, fine-tuned HuggingFace checkpoints, OpenAI and Google Gemini

AWARDS & COMPETITIVE PROGRAMMING

- **SC26 Student Volunteer** — selected with registration, airfare and accommodation covered; **1st Runner-up**, KonaSL Coding Combat 2023; **Best Performer (2016 batch)** and 11th of 90, Intra-KUET Programming Contest 2020; Rank **10 of 73** teams, Intra-KUET Programming Contest 2019
- **Codeforces: Expert**, max rating **1629**, 550+ problems over 145+ contests | **LeetCode: Knight**, max rating **1895** (**global top 5%**), 210+ problems | **CodeChef: 4-star**, max rating 1819 | **UVa: 140+** | **1100+** problems solved overall
- Reviewer, *Journal of Autonomous Intelligence* (Frontier Scientific Publishing Pte. Ltd.); Deep Learning Specialization, Coursera

LEADERSHIP & SERVICE

- **Assistant Organizing Secretary**, KUET Math Club (2020–2022) — organized the departmental math festival and ran an olympiad training workshop for 30+ participants
- **Executive Member**, SGIPC — Special Group Interested in Programming Contest, KUET (2018–2022) — ran competitive-programming onboarding workshops for 60+ first-year students and arranged online and onsite contests
- **Executive Member**, Organization of KUET Sports (2020–2022) — organized annual sporting events; captained the CSE departmental cricket team